Renal cell carcinoma

Management by **histology (clear cell vs non-clear cell)** and **IMDC risk**. Localized disease is surgical ± adjuvant pembrolizumab; metastatic clear cell gets an **IO backbone** (IO/TKI or ipi/nivo) with no predictive biomarker; **after IO progression, switch mechanism; IO rechallenge not beneficial (CONTACT-03)**

Compiled from the Stanford Oncology Handbook 2025, GU: Renal Cell Carcinoma (Ruiz SM, Fan AC) · NCCN · primary trial reports and FDA actions verified against PubMed, September 2026.

WORKUP

- **Most are found incidentally** on imaging; the classic triad (flank pain, hematuria, mass) is seen in only ~10%
- **Paraneoplastic:** erythrocytosis (EPO), **hypercalcemia** (PTHrP), hypertension (renin), **Stauffer syndrome** (cholestasis without liver mets — resolves after nephrectomy)
- **CT abdomen/pelvis with contrast** (MRI if poor renal function, IVC thrombus, or to call fat in an angiomyolipoma) + **chest CT**. Brain MRI and bone scan when symptomatic or before systemic therapy in high-risk disease
- **Biopsy is not required before nephrectomy** for a solid enhancing mass — get it for small masses when it would change AS vs treatment, suspected lymphoma/metastasis/infection, or before systemic therapy without surgery
- Labs that feed IMDC: **CBC with differential, calcium**, plus LDH, creatinine, LFTs
- **Genetics referral (NCCN):** RCC at **age ≤46**, bilateral or multifocal tumors, ≥1 first/second-degree relative with RCC, or a somatic hit with germline implications

STAGING (AJCC 8TH)

T1	Confined to kidney, **≤7 cm** — T1a ≤4 cm, T1b >4–7 cm
T2	Confined to kidney, **>7 cm** — T2a ≤10 cm, T2b >10 cm
T3	Major veins or perinephric fat, not beyond Gerota's — **a** renal vein / sinus or perirenal fat · **b** IVC below diaphragm · **c** IVC above diaphragm or caval wall
T4	Beyond Gerota's fascia, incl. contiguous **ipsilateral adrenal**
Stage	**I** T1N0 · **II** T2N0 · **III** T1–2N1 or T3 any N · **IV** T4 or M1

Grade (WHO/ISUP) is nucleolar: G1–G3 by nucleolar prominence at 400×/100×; **G4 = marked pleomorphism, giant cells, rhabdoid or sarcomatoid**.

IMDC RISK

KPS <80%	**Hemoglobin < LLN**
Diagnosis → systemic tx <1 year	**Calcium > ULN** (corrected)
Neutrophils > ULN	**Platelets > ULN**

Favorable = 0 · Intermediate = 1–2 · Poor = ≥3. Two clinical + four labs. **Prognostic, and it is how the 1L trials were stratified** — but it does not predict IO vs TKI benefit except that ipi/nivo underperforms in favorable risk.

AFTER NEPHRECTOMY

- **Surveillance** (NCCN, risk-adapted): history, labs and chest/abdominal imaging every 3–6 months for ~3 years after stage II–III, then annually to year 5 and beyond — **RCC recurs late**, sometimes a decade or more after surgery
- **Unusual metastatic sites are classic:** pancreas, thyroid, adrenal, skin, and hypervascular brain lesions that bleed

TREATMENT PATHWAY

Localized — stage I–III

Resect (or surveil the small mass)
T1a: **partial nephrectomy**, ablation <3 cm, or AS
T1b–T3: partial if feasible, else radical
renal vein / IVC thrombus: still resectable

Adjuvant — clear cell, increased risk
pT2 G4/sarcomatoid, ≥pT3, N+, or M1 NED
pembrolizumab ×1 yr (KEYNOTE-564, OS HR 0.62)
or pembro + belzutifan (LITESPARK-022, FDA 6/2026)
non-clear cell: surveillance — no adjuvant data

Metastatic clear cell — first line by IMDC

Favorable (0 factors)
IO/TKI: pembro + axitinib, nivo + cabozantinib, or lenvatinib + pembro
not ipi/nivo (ORR 30% vs 52% for sunitinib)
indolent, low-volume: active surveillance

Intermediate (1–2) / poor (≥3)
ipi/nivo (8-yr OS HR 0.69) **or any IO/TKI**
sarcomatoid: IO combination, not TKI alone
IO-ineligible: **cabozantinib** (CABOSUN)
no triplet — COSMIC-313 had no OS gain

Cytoreductive nephrectomy
selected only — usually deferred after response

Later lines & non-clear cell

After IO (1L or adjuvant)
switch to a VEGF TKI: cabozantinib, axitinib, lenvatinib + everolimus; tivozanib after ≥2 lines
belzutifan after IO + VEGF TKI
do not add IO back (CONTACT-03, TiNivo-2)
belz + lenvatinib: FDA decision due 4 Oct 2026

Non-clear cell — trial preferred
papillary: **cabozantinib** (PAPMET)
lenva + pembro or cabo + nivo (single-arm)
collecting duct / medullary: **gem + platinum**
HLRCC: **erlotinib + bevacizumab**

HISTOLOGY

Subtype	Biology · what it means
Clear cell *~75%*	**VHL loss (3p) → HIF-2α accumulation → VEGF**. Why VEGF TKIs, IO and belzutifan work; chemo-resistant
Papillary *~10–15%*	MET-driven subset (chr 7 gain, MET alterations). **Cabozantinib** best randomized data
Chromophobe *~5%*	Usually indolent; **IO barely works**. BHD (FLCN) association
Collecting duct *<1%*	Aggressive, often metastatic at diagnosis — **treat like urothelial: gemcitabine + platinum**
Renal medullary *<1%*	**Young patients with sickle cell trait**; SMARCB1 (INI1) loss. Platinum-based chemo; dismal prognosis
FH-deficient	HLRCC-associated (formerly 'type 2 papillary'); **metastasizes when small**
Translocation	TFE3/TFEB (MiT family); children and young adults
Sarcomatoid *any subtype*	Not a subtype but dedifferentiation — **poor prognosis, yet the best responders to IO**

LOCALIZED DISEASE & SURGERY

- **Small renal mass (≤4 cm): active surveillance is legitimate**, especially in the elderly or comorbid — many are benign or indolent. Serial imaging; intervene for growth (~>5 mm/yr) or size >3–4 cm. **Ablation** (cryo/RFA) for <3 cm
- **Partial nephrectomy preferred for T1a** (and T1b when feasible) to protect renal function.
- **IVC tumor thrombus is not a contraindication** to resection. Lymphadenectomy has no proven OS benefit; no role for adjuvant RT
- **Cytoreductive nephrectomy is no longer routine:** CARMENA — sunitinib alone non-inferior (OS 18.4 vs 13.9 mo); SURTIME — deferred CN after 3 cycles of sunitinib gave longer OS (exploratory). Keep it for good PS, low-volume disease, symptomatic primary, or a responder with residual primary
- **Metastasectomy / SBRT** for oligometastatic disease, especially with a long disease-free interval

PEARLS

- **No chemotherapy for clear cell** — the only chemo-sensitive RCCs are collecting duct and medullary
- **Renal mass + sickle cell trait in a young Black patient = medullary carcinoma** until proven otherwise
- **Belzutifan is the VHL drug** (hereditary) **and** a later-line drug in sporadic ccRCC — same target, HIF-2α
- **Cabozantinib appears at every step:** 1L (with nivo or alone), 2L after IO (METEOR, CONTACT-03 control arm), and papillary (PAPMET)

Renal cell carcinoma

ADJUVANT THERAPY

Trial	Agent · result
KEYNOTE-564	**Pembrolizumab ×1 yr vs placebo** — DFS HR 0.68; **OS HR 0.62** (48-mo OS 91.2% vs 86.0%). The first adjuvant therapy in RCC with an OS benefit
LITESPARK-022	**Pembro + belzutifan vs pembro** — DFS HR 0.72, 24-mo DFS 80.7% vs 73.7%. **OS immature** (HR 0.78, NS); grade ≥3 AEs **52% vs 30%**. FDA approved 12 Jun 2026

Clear cell only. Every other adjuvant IO trial (nivo ± ipi, atezolizumab, perioperative nivo) was negative, and no adjuvant TKI improved survival (sunitinib, S-TRAC, gained DFS only and is rarely used) — **pembrolizumab-based therapy is the adjuvant option**.

FIRST-LINE CLEAR CELL — ALL VS SUNITINIB UNLESS STATED

Regimen · trial	Long-term result
Ipi + nivo *CheckMate 214*	**Int/poor:** OS HR 0.69 at 8 yr, CR 11.8% vs 2.6%, median DOR 82.8 mo — the most durable tail. **Favorable:** OS HR 0.82 (NS), ORR 30% vs 52%. PFS co-primary missed its α
Pembro + axitinib *KEYNOTE-426*	Final (43 mo): **OS HR 0.73**, PFS HR 0.68, ORR 60% vs 40%. All risk groups
Nivo + cabozantinib *CheckMate 9ER*	Final (5.6 yr): **OS 46.5 vs 35.5 mo, HR 0.79**; PFS 16.4 vs 8.3 mo; ORR 55.7% vs 27.4%, CR 13.9%. Best QoL data
Lenvatinib + pembro *CLEAR*	**PFS 23.9 vs 9.2 mo** (HR 0.47), ORR 71.3% — highest response rate. Final OS HR 0.79, **medians equal** (53.7 vs 54.3 mo)
Cabo + nivo + ipi *COSMIC-313*	vs ipi/nivo, int/poor — PFS 16.6 vs 11.2 mo (HR 0.82) but **OS HR 1.02**, grade 3–4 AEs 79% vs 56%. **Triplet rejected**
Cabozantinib *CABOSUN*	Int/poor, **randomized phase 2** — PFS 8.2 vs 5.6 mo. The IO-ineligible option

No head-to-head trial between IO combinations and no predictive biomarker (PD-L1 does not select). Pick by risk group, comorbidity, and whether you want response (IO/TKI) or a durable treatment-free tail (ipi/nivo).

FIRST-LINE REGIMEN CHOICE

- **Symptomatic, bulky, or need a response now** → IO/TKI; **lenvatinib + pembro** has the highest ORR and longest PFS
- **Int/poor risk, want a chance at a durable treatment-free remission** → **ipi/nivo**; also avoids chronic TKI toxicity. Expect front-loaded irAEs
- **Favorable risk** → IO/TKI (ipi/nivo lost on ORR and PFS); indolent low-volume disease can be watched
- **Autoimmune disease, transplant, or IO-ineligible** → cabozantinib (int/poor) or sunitinib/pazopanib
- **Relapse after adjuvant pembrolizumab:** no randomized data — most use a VEGF TKI-based regimen, especially if relapse occurs on or soon after IO

SECOND LINE AND BEYOND

Trial	Setting · result
LITESPARK-011	**Belzutifan + lenvatinib vs cabozantinib** after anti-PD-(L)1 — PFS 14.8 vs 10.7 mo (HR 0.70), ORR ~53% vs ~40%; **interim OS HR 0.85, not significant** (final pending). FDA action date 4 Oct 2026
LITESPARK-005	**Belzutifan vs everolimus** after IO + TKI — ORR 21.9% vs 3.5%, 18-mo PFS 24% vs 8%, but **median PFS 5.6 mo in both** and **OS HR 0.88, NS**. FDA 14 Dec 2023
METEOR	**Cabozantinib vs everolimus** after VEGF TKI — OS 21.4 vs 16.5 mo (HR 0.66), PFS HR 0.51
CheckMate 025	**Nivolumab vs everolimus** after VEGF TKI — OS 25.0 vs 19.6 mo (HR 0.73); the IO-naive 2L option
Study 205	**Lenvatinib + everolimus** vs everolimus, randomized phase 2 — PFS 14.6 vs 5.5 mo (HR 0.40)
TIVO-3	**Tivozanib vs sorafenib** after ≥2 lines — PFS 5.6 vs 3.9 mo; **no OS benefit**
CONTACT-03 · TiNivo-2	Adding atezolizumab or nivolumab to a TKI after IO — **no benefit**

Do not rechallenge IO. After IO, the next drug is a VEGF TKI; belzutifan comes after IO + TKI.

NON-CLEAR CELL — MOSTLY SINGLE-ARM DATA

Histology / trial	Evidence
Papillary *SWOG S1500 (PAPMET)*	**Cabozantinib vs sunitinib**, randomized phase 2 — PFS 9.0 vs 5.6 mo (HR 0.60), ORR 23% vs 4%; OS NS. Savolitinib and crizotinib arms closed for futility
Mixed nccRCC *KEYNOTE-B61*	**Lenvatinib + pembro**, single-arm — ORR 49% (51% at 2 yr), sarcomatoid 47%
Mixed nccRCC *phase 2*	**Cabo + nivo** — ORR 47.5% papillary/unclassified; **0 of 7 chromophobe responded**
Chromophobe	IO largely inactive; VEGF TKI ± mTOR (e.g., lenvatinib + everolimus). Trial preferred
Collecting duct · medullary	**Gemcitabine + cisplatin/carboplatin**; TKIs and IO rarely help. **Check sickle status in a young patient with a renal mass**
Sarcomatoid	Use an **IO combination** whatever the parent histology — the highest CR rates in 1L subgroups

HEREDITARY SYNDROMES (ALL AUTOSOMAL DOMINANT)

Syndrome / gene	Renal tumor · other features · management
VHL *VHL, 3p*	**Clear cell, multifocal/bilateral**; CNS/retinal hemangioblastomas, pheochromocytoma, pNET, pancreatic cysts. **Belzutifan** (LITESPARK-004: RCC ORR 49%; FDA Aug 2021) spares repeated surgery; resect when the largest tumor reaches ~3 cm
HLRCC *FH*	**FH-deficient RCC** — solitary, aggressive, metastasizes early: **excise promptly, no surveillance**. Cutaneous + uterine leiomyomas. Advanced: **erlotinib + bevacizumab** (ORR 72%, mPFS 21.1 mo)
Birt-Hogg-Dubé *FLCN*	**Chromophobe / hybrid oncocytic**, bilateral; fibrofolliculomas, lung cysts → pneumothorax
Tuberous sclerosis *TSC1/2*	**Angiomyolipomas** (rarely RCC); **everolimus**. Angiofibromas, SEGA, LAM
BAP1 TPDS *BAP1*	Aggressive **clear cell**; uveal and cutaneous melanoma, mesothelioma
HPRC *MET*	**Papillary**, multifocal and bilateral
Hereditary PGL/PCC *SDHB/C/D*	**SDH-deficient RCC**; paraganglioma, pheochromocytoma, GIST

Renal cell carcinoma

REGIMEN DOSING

Regimen	Dosing
Adjuvant pembrolizumab *KEYNOTE-564*	**200 mg IV q3wk ×17** or 400 mg q6wk (~1 yr)
+ Belzutifan *LITESPARK-022*	Belzutifan **120 mg PO daily ×up to 54 wk** with pembrolizumab 200 q3wk / 400 q6wk ×12 mo
Pembro + axitinib	Axitinib **5 mg PO BID** (titrate to 7 then 10 mg BID if BP controlled) + pembrolizumab 200 mg q3wk / 400 q6wk, IO up to 2 yr
Ipi + nivo	Nivolumab **3 mg/kg** + ipilimumab **1 mg/kg** IV **q3wk ×4** → nivolumab 240 mg q2wk or 480 mg q4wk
Nivo + cabozantinib	Cabozantinib **40 mg** PO daily (not 60) + nivolumab 240 q2wk / 480 q4wk, nivo max 2 yr
Lenvatinib + pembro	Lenvatinib **20 mg** PO daily + pembrolizumab 200 q3wk / 400 q6wk, IO up to 2 yr
Cabozantinib	**60 mg PO daily** on an empty stomach; hold ≥3 wk before surgery
Belzutifan	**120 mg PO daily** — after IO + VEGF TKI, and for VHL tumors
Belzutifan + lenvatinib	120 mg + **20 mg** daily (LITESPARK-011; pending FDA)
Tivozanib	**1.34 mg PO daily, 21 days on / 7 off** — after ≥2 prior systemic lines
Lenvatinib + everolimus	Lenvatinib **18 mg** + everolimus **5 mg** PO daily
Axitinib alone	5 mg BID → 7 → 10 mg BID if BP ≤150/90 and tolerated
Sunitinib	**50 mg PO daily, 4 wk on / 2 wk off** (or 2 on / 1 off for tolerance)
Pazopanib	**800 mg PO daily**, empty stomach
Nivolumab alone	240 mg q2wk or 480 mg q4wk
Erlotinib + bevacizumab	Erlotinib 150 mg PO daily + bevacizumab 10 mg/kg IV q2wk — HLRCC / papillary

Pembrolizumab in RCC is 200 mg q3wk or 400 mg q6wk — not 300/600. Historical: high-dose IL-2 (ORR ~14%, rare durable CRs), temsirolimus (poor-risk, ARCC).

TOXICITY & MONITORING

- **Hypertension is the signature VEGF-TKI toxicity** — control BP before starting, home BP log, add agents early rather than stopping the TKI. Urine protein with lenvatinib and bevacizumab
- **Transaminitis on axitinib + pembrolizumab** (or cabo/lenva + IO): **hold both drugs**. TKI hepatitis resolves within days (short half-lives); **if ALT is not falling after ~1 week, or grade ≥3, treat as immune hepatitis** with prednisone 1–2 mg/kg. Restart the TKI first at reduced dose; rechallenge IO only if it wasn't the culprit
- **Pazopanib carries a boxed warning for hepatotoxicity** — LFTs before starting and every ~2 weeks through month 4; also QT prolongation
- **Hypothyroidism** from both TKIs (sunitinib, axitinib, lenvatinib) and IO — TSH every 4–6 wk on treatment
- **Belzutifan: on-target anemia** (HIF-2α drives EPO — decreased Hgb is the most common adverse reaction) and **hypoxia**: check Hgb and pulse oximetry at baseline and periodically; transfuse, hold or dose-reduce. **Boxed warning: embryo-fetal toxicity**, and it can make hormonal contraception ineffective
- **Ipi/nivo:** front-loaded irAEs — colitis, hepatitis, hypophysitis, adrenal insufficiency. Check cortisol with fatigue or hyponatremia
- **Cabozantinib:** diarrhea, hand-foot, stomatitis, wound-healing and bleeding — dose reductions to 40/20 mg are routine
- **Sunitinib:** hand-foot, mucositis, cytopenias, cardiomyopathy. **mTOR inhibitors:** pneumonitis, hyperglycemia, hyperlipidemia
- **Bone metastases** (lytic, ~1/3): zoledronic acid (renally dosed) or denosumab; **brain metastases** — SRS before or alongside systemic therapy

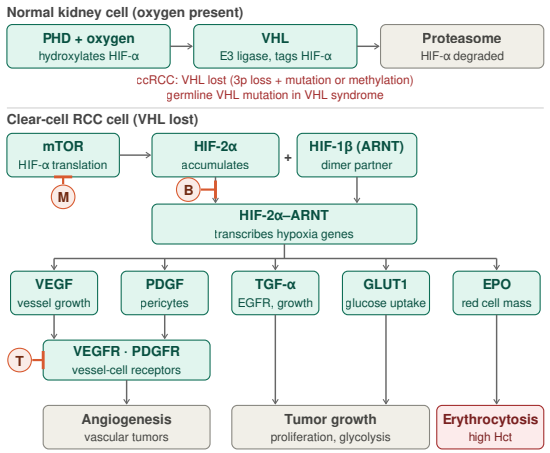
SPECIAL SITUATIONS

- **Brain metastases:** treat locally first (SRS or resection); patients with treated, stable brain mets can receive standard systemic therapy
- **Dialysis / poor renal function:** IO needs no renal adjustment; dose-adjust zoledronic acid, and **denosumab carries a severe-hypocalcemia risk in advanced CKD**
- **Pre-existing autoimmune disease or solid-organ transplant:** avoid IO; TKI monotherapy
- **Surgery on a TKI:** hold cabozantinib ≥3 wk before and resume only after wound healing; bevacizumab ≥28 days

KEY TRIALS

KEYNOTE-564	Adjuvant pembrolizumab — DFS HR 0.68, **OS HR 0.62**
LITESPARK-022	Adding belzutifan to adjuvant pembro — DFS HR 0.72, OS immature
CheckMate 214	Ipi/nivo, int/poor — 8-yr OS HR 0.69, durable CRs
CLEAR	Lenva + pembro — PFS 23.9 vs 9.2 mo, ORR 71%
COSMIC-313	Triplet cabo + ipi/nivo — PFS gain, **no OS gain**
CARMENA · SURTIME	Immediate cytoreductive nephrectomy not needed; defer and select
METEOR	Cabozantinib after TKI — OS HR 0.66
LITESPARK-005 / -011	Belzutifan after IO + TKI (ORR 22%); + lenvatinib post-IO, PFS HR 0.70
CONTACT-03 · TiNivo-2	IO rechallenge after IO — **no benefit**
PAPMET	Papillary: cabozantinib beats sunitinib on PFS
LITESPARK-004	Belzutifan in VHL disease — RCC ORR 49%
KEYNOTE-426 · 9ER	IO/TKI vs sunitinib — final OS HR 0.73 / 0.79
CheckMate 025	Nivolumab after TKI — OS 25.0 vs 19.6 mo
NCI bev + erlotinib	HLRCC papillary RCC — ORR 72%, mPFS 21.1 mo

PATHOPHYSIOLOGY



VHL loss mimics hypoxia: HIF-2α stays on and drives VEGF, so VEGF TKIs work. Belzutifan blocks HIF-2α upstream; it also lowers EPO, so anemia is on-target.